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HSPT 161 02 - Wines of the World

Project Summary - Density and Temperature Based Conceptual ABV Calculator for the Arduino

Process:

When looking for topics to research for the project I focused primarily on computer engineering tools that wine making uses, which I discovered was surprisingly few. A lot of the engineering for wine making goes into the automation of wine making, which I had no particular interest in. And one day during class we once again said “I wouldn’t be surprised if this was on the upper end of the 1.5% alcohol tolerance”, and so I wanted to research making a device so we wouldn’t have to wonder. A big part of my topic was that I already knew the basics of what I wanted to use. An arduino is a cheap and effective way to make this sort of mini-project but the tools I needed, so that was obvious, but the additional parts were the key factor I needed.

Research Foundation:

My original design for the project was an alcohol vapor based identifier, with my design replacing the cork of a wine bottle and reading the alcohol that way. But that design had to be scrapped as the vapor sensors aren’t good for this application. They typically require a high flow rate (like someone blowing into a breathalyzer) and can be easily inaccurate due to sugar content of wine, how long it’s been sitting, or not sitting if bubbles are present, and how recently the bottle had been opened. So I pivoted to looking into liquid alcohol sensors, which also came up with poor results. A lot of liquid alcohol sensors are larger, clunkier, more expensive, and require a high flow rate like pouring. While I could have made these work I wanted to keep the vision of “put the device in, and it does it’s job”. As such I ended up going deeper into how to calculate

ABV rather than how to read it. Water has a standard density of 1g/cm³, but wine has a density below that of water. In fact, wine typically has a density that decreases based on alcohol percentage. And this fact has a formula...

$$ABV (\%) = (OG - FG) * 131.25$$

Where OG is the original gravity (pre-fermentation) and FG is the final gravity (post-fermentation). But as you may notice this equation has a fatal flaw for my purposes, OG and FG are pre and post fermentation. And it would be wrong to assume the post fermentation density is 1g/cm³ like water, making this equation unusable for me.

So the only thing left I could do was follow trends, initially I wanted to static define constant values to each density. So 0.996 would be 0% ABV, 0.966 would be 20% ABV, or something similar. But the obvious issue here is that density will not always fall into those static brackets, so I needed something that could read between normal values. The best way to do this is a trend line, and reverse engineer an equation from there.

But I ran into a problem, there is no standard for ABV by densities for wines, and even so a wine's density is composed of not just grapes and alcohol, but other components of the wine like residual sugars. So I based the model not on wine, but on ethanol water densities, using a table from the International Organisation of Vine and Wine which highlights alcohol volume by density at 20 degrees celsius. With this table I resulted with...

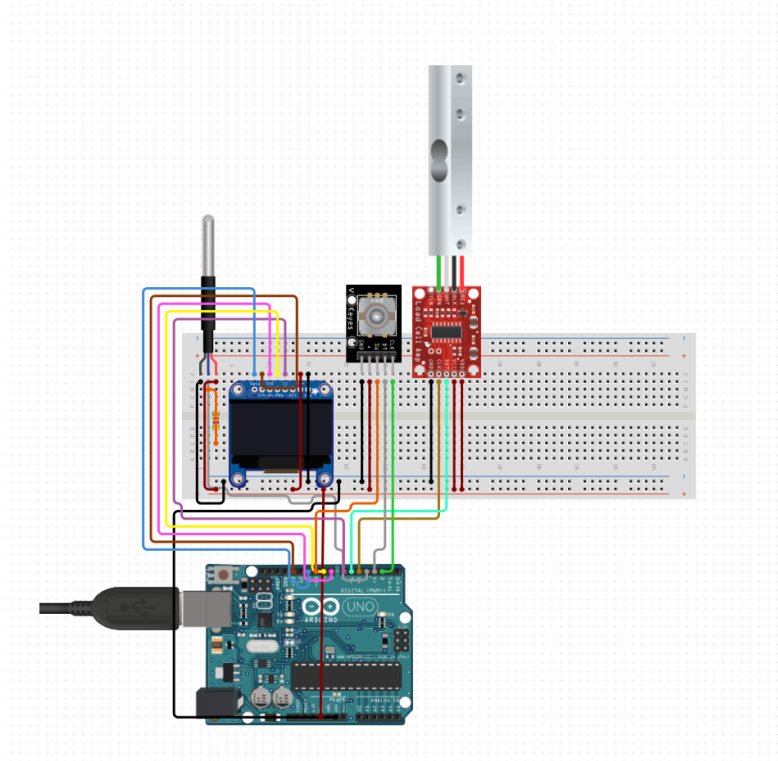
$$abv = 19399.7x^2 + 642.6x - 0.050$$

Where d is the temperature compensated density measurement from the load cell. With this I can semi-accurately predict the ABV based on the density and temperature of the liquid measured using my prototype.

Conclusion:

What I learned from this project was that wine, even from a technical standpoint like engineering, is significantly more complicated than you might think. When I initially began research for my design, I thought it would be relatively simple. It turns out that was as far from the truth as could be, and even now there is still

significant room for improvement in my design. While the water-ethanol based empirical equation works well, it still leaves out sugars, tannins, and other factors of wine that can affect density. Especially when I learned that winemakers use the pre and post fermentation densities to accurately measure ABV, because the fermentation process changes the wine so much having both values is not just helpful but necessary. I picked this topic because engineering a device fit well with my skills and interests, but it happily ended up leading to much more in depth learning about how these devices work, and how similar devices are used in the process of making wine. I believe that if this were a capstone project, I could make an incredibly accurate ABV measuring device with what I've learned.



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